# 23 Lecture 23 (March 10, 2025, est.): Beyond fusion categories

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This final lecture describes the extension of the Frobenius method to nonsemisimple categories, and the role of higher Verlinde categories. Throughout, $k$ is algebraically closed of characteristic $p > 0$, and $\mathcal{C}$ is a symmetric tensor category with finite-dimensional Hom-spaces and finite-length objects. Recall that *moderate growth* means: for every $X$ there is a real constant $d_X$ such that $\ell(X^{\otimes n}) \leq d_X^n$ for every $n \geq 1$. The results and conjectures below retain the distinction made in the course between proved structure theorems and the proposed higher theory.

## 23.1 Frobenius functors in the general case

For $G \leq S_n$, the symmetry gives a functor

$$P_G : \mathcal{C} \longrightarrow \mathrm{Rep}(G, \mathcal{C}), \qquad X \longmapsto X^{\otimes n}$$

with the restricted permutation action. There is a general construction [CEO23, Section 3.1]

$$\mathrm{Fr}_G : \mathcal{C} \longrightarrow \mathcal{C} \boxtimes \overline{\mathrm{Rep}_k(G)}. \tag{23.1}$$

Its second step applies semisimplification in the $G$-representation variable. When $\mathcal{C}$ is not semisimple, this needs care: $\mathrm{Rep}(G, \mathcal{C})$ is not generally $\mathcal{C} \boxtimes \mathrm{Rep}_k(G)$, and quotienting *all* negligible morphisms of $\mathrm{Rep}(G, \mathcal{C})$ would also lose information from $\mathcal{C}$.

One precise construction presents $\mathcal{C}$ as comodules of a cotriangular copseudo-Hopf algebra $H$. Such a presentation records the abelian category by a coalgebra and its tensor structure by compatible bicomodule data. Then

$$\mathrm{Rep}(G, \mathcal{C}) \simeq H\text{-Comod}(\mathrm{Rep}_k(G)).$$

Apply $\mathrm{Rep}_k(G) \to \overline{\mathrm{Rep}_k(G)}$ to the representation-variable data and the coactions. The result is $H\text{-Comod}(\overline{\mathrm{Rep}_k(G)}) \simeq \mathcal{C} \boxtimes \overline{\mathrm{Rep}_k(G)}$. This construction is independent of the chosen presentation, and is symmetric monoidal; these facts are part of the cited construction.

The functor (23.1) is additive precisely when a Sylow $p$-subgroup of $G$ acts transitively on $\{1, \ldots, n\}$. In particular, additivity requires $n = p^j$. In this case it is $p^j$-semilinear and becomes $k$-linear after replacing the first factor of its target by $\mathcal{C}^{(j)}$, with scalar action $\lambda \cdot f = \lambda^{p^j} f$.

The principal examples are

$$\mathrm{Fr} = \mathrm{Fr}_{C_p} : \mathcal{C} \longrightarrow \mathcal{C}^{(1)} \boxtimes \mathsf{Ver}_p$$

and the *enhanced Frobenius functor*

$$\mathrm{Fr}^{\mathrm{en}} = \mathrm{Fr}_{S_p} : \mathcal{C} \longrightarrow \mathcal{C}^{(1)} \boxtimes \overline{\mathrm{Rep}_k(S_p)}.$$

Neither is assumed exact in a general category.

## 23.2 Invariants, coinvariants, and the functor $\mathrm{Fr}_+$

For $m \geq 1$, write

$$\Gamma^m X = (X^{\otimes m})^{S_m}, \qquad S^m X = (X^{\otimes m})_{S_m}.$$

These are respectively the largest subobject on which all permutations act trivially and the largest quotient with trivial permutation action. The inclusion followed by the quotient gives a canonical map $\Gamma^m X \to S^m X$.

**23.1 Definition.** For $j \geq 1$, define

$$\mathrm{Fr}_+^{(j)}(X) = \mathrm{im}\big(\Gamma^{p^j} X \longrightarrow S^{p^j} X\big), \qquad \mathrm{Fr}_+ = \mathrm{Fr}_+^{(1)}.$$

These are $p^j$-semilinear functors. The functor $\mathrm{Fr}_+$ is a direct summand of $\mathrm{Fr}^{\mathrm{en}}$, extracted by the unit multiplicity in $\overline{\mathrm{Rep}_k(S_p)}$. It has a canonical lax monoidal structure; its structural maps are not asserted to be isomorphisms.

**23.2 Example.** For an ordinary vector space $V$ with basis $v_1, \ldots, v_d$, the image of $\Gamma^{p^j} V \to S^{p^j} V$ has basis $v_1^{p^j}, \ldots, v_d^{p^j}$. Indeed, the orbit sum of a mixed tensor maps to its orbit size times the corresponding monomial. The multinomial coefficient for a nonconstant word of length $p^j$ is divisible by $p$, whereas constant words have orbit size one. Thus this functor recovers the usual Frobenius twist of a vector space, and similarly the Frobenius twist of a representation.

**23.3 Theorem** (Coulembier–Etingof–Ostrik)**.** *The following conditions are equivalent:*

1. $\mathrm{Fr}$ *is exact;*
2. $\mathrm{Fr}^{\mathrm{en}}$ *is exact;*
3. $\mathrm{Fr}_+$ *is exact.*

*A category satisfying them is called* Frobenius exact*.*

See [CEO23, Proposition 3.4]. The equivalence is a theorem, not a formal consequence of additivity. In particular, semisimple categories are Frobenius exact, but nonsemisimple examples need not be.

**23.4 Theorem** (The Verlinde fiber-functor criterion)**.** *The category $\mathcal{C}$ is Frobenius exact and of moderate growth if and only if it admits an exact $k$-linear symmetric monoidal functor $\mathcal{C} \to \mathsf{Ver}_p$.*

This is the general structure theorem of [CEO23, Theorem 7.13]. The forward implication is a deep extension of the fusion-category argument. For the reverse implication, exact faithful functors preserve a strict composition series and hence cannot decrease length. Tensor powers in the fusion category $\mathsf{Ver}_p$ have exponentially bounded length, proving moderate growth. Naturality of the Frobenius construction and exact faithfulness of the fiber functor then detect exactness of $\mathrm{Fr}$, since the target is semisimple.

**23.5 Lemma** (Iteration)**.** *If $\mathcal{C}$ is Frobenius exact, then*

$$\mathrm{Fr}_+^{(j)} \simeq (\mathrm{Fr}_+)^{\circ j} \qquad (j \geq 1).$$

This is [CEO23, Lemma 5.1]; the equality compares the same semilinearity on both sides. Its proof uses a faithful exact symmetric functor that splits all short exact sequences, and the comparison of invariants and coinvariants for iterated permutation groups. Without Frobenius exactness the assertion can fail. Thus the definition using $S_{p^j}$-invariants must not be replaced by iteration without this hypothesis.

## 23.3 Higher Verlinde categories and incompressibility

**23.6 Definition.** A symmetric tensor category $\mathcal{D}$ is *incompressible* if every tensor functor from $\mathcal{D}$ to another symmetric tensor category is an embedding of a tensor subcategory: it is fully faithful and its image is closed under subobjects and quotients.

Besides Vect, SVec for $p > 2$, and $\mathsf{Ver}_p$, there are finite symmetric tensor categories $\mathsf{Ver}_{p^n}$ and $\mathsf{Ver}_{p^n}^+$ for $n \geq 1$, constructed from quotients of the category of tilting modules for $\mathrm{SL}_2$ and their abelian envelopes [BEO20]. There are tensor embeddings

$$\mathsf{Ver}_p \subseteq \mathsf{Ver}_{p^2} \subseteq \mathsf{Ver}_{p^3} \subseteq \cdots .$$

These categories and their indicated subcategories are incompressible. For $n > 1$, $\mathsf{Ver}_{p^n}$ is not Frobenius exact. In particular, moderate growth alone does not imply the existence of a fiber functor to $\mathsf{Ver}_p$.

**23.7 Theorem** (Coulembier–Etingof–Ostrik)**.** *Every symmetric tensor category of moderate growth admits a tensor functor to an incompressible symmetric tensor category.*

This existence theorem is proved in [CEO24]. It leaves the classification of possible incompressible targets as a separate problem. The conjecture discussed in the course is that every category of moderate growth fibers over $\mathsf{Ver}_{p^\infty} = \bigcup_n \mathsf{Ver}_{p^n}$; the known higher Verlinde categories and their tensor subcategories would then supply all the incompressible targets in this class. This is stated as a conjecture, not used as a theorem.

## 23.4 A proposed higher Frobenius construction

We conclude with the construction of Coulembier–Flake [CF24]. Fix $n \geq 1$ and $E = (C_p)^n$. Choose a faithful two-dimensional representation $V$ of $E$, equivalently an embedding $E \hookrightarrow \mathrm{SL}_2(k)$. For example, choose $\mathbb{F}_p$-linearly independent elements $a_1, \ldots, a_n \in k$ and send the $i$th generator to $\left(\begin{smallmatrix} 1 & a_i \\ 0 & 1 \end{smallmatrix}\right)$. Let $\mathcal{D}$ be the full additive idempotent-complete monoidal subcategory generated by $V$ in $\mathrm{Rep}_k(E)$.

There is a tensor ideal of morphisms $\mathcal{K}$ in $\mathcal{D}$ such that $\mathsf{Ver}_{p^n}$ is the monoidal abelian envelope of $\mathcal{D}/\mathcal{K}$. This is an actual construction theorem. For $n = 1$ it recovers the

quotient of $\mathrm{Rep}_k(C_p)$ by negligible maps. Together with the regular permutation embedding $E \hookrightarrow S_{p^n}$, it leads to candidate functors

$$\Phi_{\mathcal{C}}^{V} : \mathcal{C} \longrightarrow \mathcal{C}^{(n)} \boxtimes \mathsf{Ver}_{p^n},$$

based on the $E$-action on $X^{\otimes p^n}$. The exponent in the Frobenius scalar twist is $p^n$.

The category $\mathcal{D}$ has only finitely many indecomposables and has a smallest nonzero *thick tensor ideal of objects* $\mathcal{J}$. Here "thick" means closed under finite direct sums and direct summands; being a tensor ideal means closure under tensoring with every object of the ambient category. This ideal of objects is distinct from the ideal of morphisms $\mathcal{K}$ above.

The two conjectural steps discussed in the handwritten notes are:

1. The candidate functors $\Phi_{\mathcal{C}}^{V}$ admit compatible symmetric monoidal structures and intertwine tensor functors in $\mathcal{C}$, giving higher analogues of the ordinary Frobenius functor.
2. The full subcategory $\mathcal{J} \subseteq \mathcal{D} \subseteq \mathrm{Rep}_k(E)$ remains a thick tensor ideal in the entire category $\mathrm{Rep}_k(E)$, including closure under tensoring with modules outside $\mathcal{D}$.

Coulembier–Flake prove that the second statement implies the first. They establish special cases, but the general monoidality assertion is not a consequence merely of taking a tensor power. The longer-term aim is an exactness criterion for fibering over $\mathsf{Ver}_{p^n}$, parallel to Theorem 23.4 when $n = 1$.

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